

Task 2 — Observability Deep-Dive, SLOs & Error Budgets

Data Analyst • Phase 3 • Demonstration report

Important data note

The supplied screenshots contain task briefs but no production/source dataset, event logs, or backend performance logs. The numerical evidence in this package is DEMONSTRATION data and must be replaced/reconciled with actual source data before being presented as real production evidence.

Executive summary

The metric layer gives each metric one explicit definition, named source, refresh expectation and decision use. The quality layer checks freshness, volume, nulls and uniqueness. Each test has a threshold and action so an alert leads to a defined response.

Published metric dictionary

metric	definition	source	refresh	decision_use
Visits	count(distinct session_id)	session_events	Daily	Launch traffic / funnel denominator
Application Starts	count(distinct application_id where start_event=1)	application_events	Daily	Measures intent after visit
Document Upload Success Rate	successful_uploads / upload_attempts	document_events	Hourly	Detects upload reliability issues
Submission Conversion	submissions / visits	application_events + session_events	Daily	Primary funnel outcome
Revenue	sum(successful_payment_amount)	payment_events	Daily	Business outcome
Freshness	max(event_timestamp) compared with current time	event_events	Hourly	Alerts when pipeline is stale
Null Rate	null_rows / total_rows	metric_source	Hourly	Detects missing critical values
Uniqueness Rate	unique_key_rows / total_rows	metric_source	Daily	Detects duplicate records

Data-quality tests

test	field	threshold	observed	status	action
freshness	event_timestamp	15 minutes	8 minutes	PASS	No alert
volume	daily_event_count	>= 10000	12540	PASS	No alert
null_rate	application_id	<= 1%	0.32%	PASS	No alert
uniqueness	event_id	>= 99.5%	99.92%	PASS	No alert
null_rate	revenue	<= 1%	0.18%	PASS	No alert